

Banking Data Analysis

A relational banking analytics case study using Python, Pandas, PostgreSQL, SQL and Tableau Public.

PROJECT

Banking Analytics

End-to-end portfolio project

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ANALYTICAL STACK

Python / Pandas • PostgreSQL / SQL • Tableau Public

PROJECT FOCUS

Customer and branch performance, deposits and transactions, loan and repayment risk, card fraud monitoring, and customer-support quality.

FROM DATA TO DECISIONS

The project connects relational banking data to business questions, validates metrics at the correct grain, and presents the findings through seven Tableau Public dashboard views.

Prepared for portfolio and interview presentation

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A concise route through the analytical story.

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Executive snapshot

What the project is, what it measures, and why it matters.

CUSTOMERS 60,000 source table scale	BRANCHES 150 branch network	ACCOUNTS 95,000 account base	LOANS 22,000 loan portfolio
CARDS 65,000 cards in circulation	TRANSACTIONS 2,000,000 account-level	CARD TXNS 3,000,000 card-level	SUPPORT 25,000 support tickets

The business problem

A retail bank needs a reliable way to understand customer composition, deposits, branch performance, lending exposure, repayment behavior, card spending, fraud signals and customer support quality. The project turns these relational data domains into a connected analytical workflow rather than isolated charts.

Core analytical questions

- Where is deposit and lending activity concentrated?
- Which customer segments combine meaningful income, credit exposure and repayment signals?
- Which loan types and branches show higher default / late-payment exposure?
- Where is card spending concentrated and where does fraud appear?
- Which support issues consume the most service capacity and how does satisfaction vary?

Scope

Descriptive and diagnostic analytics only. The project does not claim predictive credit scoring or machine-learning risk models.

Project framing & business questions

A business-first view of the analytical design.

Project objective

- Model a normalized banking dataset in PostgreSQL with explicit relationships.
- Use Python / Pandas for exploration, cleaning and grouped analysis.
- Answer structured business questions with PostgreSQL SQL.
- Translate findings into seven Tableau Public views for executive, operational and risk audiences.
- End with management-oriented recommendations rather than stopping at descriptive charts.

Analysis map

DOMAIN	KEY QUESTIONS	OUTPUT
KPI	Bank scale and headline measures	Executive context
Customer	Income, credit score, segmentation, product ownership	Customer 360
Branch	Deposits, customers, loans, transactions, staffing	Branch Performance
Accounts	Account mix, balances, branch deposits	Money Movement
Loans	Portfolio mix, default, late payments, outstanding principal	Loan Command Center
Cards & Fraud	Card mix, spending, merchant exposure, fraud	Fraud Command Center
Support	Issue volume, satisfaction, resolution time	Risk / support view

Naming convention used in the current SQL bundle

The report uses the exact cleaned table names visible in the supplied SQL: customers_cleaned, branches_cleaned, accounts_cleaned, employee_cleaned, loan_cleaned, loan_pay_cleaned, card_cleaned, card_trans_cleaned, tranc_cleaned and sup_tic_cleaned.

Dataset, schema & data preparation

The relational foundation behind the dashboards.

TABLE	ROWS	ROLE
branches_cleaned	150	Branch master
employee_cleaned	1,800	Employees / branch assignment
customers_cleaned	60,000	Demographics, income, credit score
accounts_cleaned	95,000	Savings / current / FD accounts
loan_cleaned	22,000	Loan contracts and status
loan_pay_cleaned	600,000	Loan payment history
card_cleaned	65,000	Debit / credit cards
card_trans_cleaned	3,000,000	Card spending + fraud flag
tranc_cleaned	2,000,000	Account-level transactions
sup_tic_cleaned	25,000	Support tickets

Relationship model

- branches → employees, accounts, loans
- customers → accounts, loans, cards, support tickets
- accounts → transactions and cards
- loans → loan_payments
- cards → card_transactions

Preparation approach

The supplied notebook shows CSV ingestion with Pandas, column-name trimming, string cleanup, numeric conversion where required and date parsing. The original project documentation characterizes the source as pre-generated synthetic data, so the main engineering focus is relational integrity, loading and analytical correctness rather than heavy missing-value imputation.

Representative Pandas preparation

```
# Clean column names
for table in tables.values():
    table.columns = (
        table.columns.str.strip().str.strip('')
    )

# Clean string values
for col in table.select_dtypes(include="str").columns:
    table[col] = table[col].str.strip().str.strip('')
```

SQL analysis architecture

PostgreSQL is the business-question engine.

Analysis coverage

Seven SQL analysis files cover KPI, customer, branch, account/deposit, loan/credit-risk, card/fraud and support analysis. The Python notebook also contains transaction-analysis work.

FILE	FOCUS	QUESTION COUNT
KPI_ANALYSIS.sql	Headline KPIs	7 KPI statements
customers analysis.sql	Customer Analysis	10
branch Performance.sql	Branch Performance	6
Account & Deposit Analysis.sql	Accounts & Deposits	8
Loan & Credit Risk.sql	Loan & Credit Risk	9
Card & Fraud Analysis.sql	Cards & Fraud	7
Customer Supports.sql	Customer Support	5

Representative SQL pattern

```
SELECT
  occupation,
  ROUND(AVG(annual_income), 2) AS average_income,
  COUNT(DISTINCT customer_id) AS customer_count
FROM customers_cleaned
GROUP BY occupation
ORDER BY average_income DESC;
```

Why this matters

- GROUP BY converts row-level records into decision-ready business segments.
- COUNT(DISTINCT ...) protects customer metrics from duplicate relationship paths.
- Window functions support ranking and percentage-style portfolio analysis.
- Independent subqueries or CTEs keep multi-table aggregations at the correct grain.

SQL quality principle

Aggregate each business domain before combining branch-level results. This prevents join fan-out from inflating balances, customer counts or transaction measures.

Python / Pandas analytical workflow

Exploration, validation and business-question prototyping.

The notebook is organized into analysis groups rather than one monolithic transformation. That makes the work easier to audit and maps naturally to the dashboard domains.

STEP	WHAT HAPPENS	EXAMPLE
1. Ingest	Read all source CSVs into DataFrames	<code>pd.read_csv(...)</code>
2. Standardize	Trim column names / string values	<code>str.strip()</code>
3. Type-cast	Numeric and date conversions	<code>pd.to_numeric</code> / <code>pd.to_datetime</code>
4. Explore	Group, aggregate and rank	<code>groupby</code> + <code>agg</code> + <code>sort_values</code>
5. Enrich	Merge branch/customer attributes	<code>merge(..., how="left")</code>
6. Segment	Create bands / thresholds	<code>pd.cut</code> / <code>quantile</code>
7. Translate	Use validated outputs in SQL/Tableau	Business-facing views

Representative loan-risk logic

```
loan_type_distribution = (
    loan.groupby("loan_type", as_index=False)
        .agg(loan_count=("loan_id", "nunique"),
            total_loan_amount=("loan_amount", "sum"),
            average_loan_amount=("loan_amount", "mean"))
)

loan_exposure_threshold = customer_risk[
    "total_loan_exposure"
].quantile(0.75)
```

Interpretation

The strongest risk questions are deliberately multi-signal: low credit score + high loan exposure + evidence of late payment. This is descriptive risk screening, not a predictive score.

Dashboard system & validation

Seven Tableau Public views form the presentation layer.

Dashboard set

- PRIMEBANK | Executive Banking Overview
- PRIMEBANK | CUSTOMER 360
- PRIMEBANK | Branch Performance
- PRIMEBANK | MONEY MOVEMENT
- PRIMEBANK | LOAN COMMAND CENTER
- PRIMEBANK | FRAUD COMMAND CENTER
- PRIMEBANK | RISK COMMAND CENTER

Validation checks

- KPI labels are checked against the field and aggregation used in the view.
- Fraud rate uses all card transactions in the denominator rather than a fraud-only filter.
- Branch, customer, account and transaction measures are reviewed at their intended grain.
- Multi-table relationships are structured to reduce duplication and aggregation fan-out.
- Dashboard titles and business descriptions follow the actual seven-view implementation.

Data-quality note

The Money Movement screenshot currently displays 95,000 as a customer-base KPI and 20,000,000 transactions. The source-table scale is 60,000 customers and 2,000,000 account-level transactions, so these two KPI mappings should be corrected in Tableau. The Branch Performance view also shows 1,800 employees in its KPI row; this is a valid network total but should be labeled as employee count.

PRIMEBANK | Executive Banking Overview

Bank-wide scale, deposits, loans and customer economics.

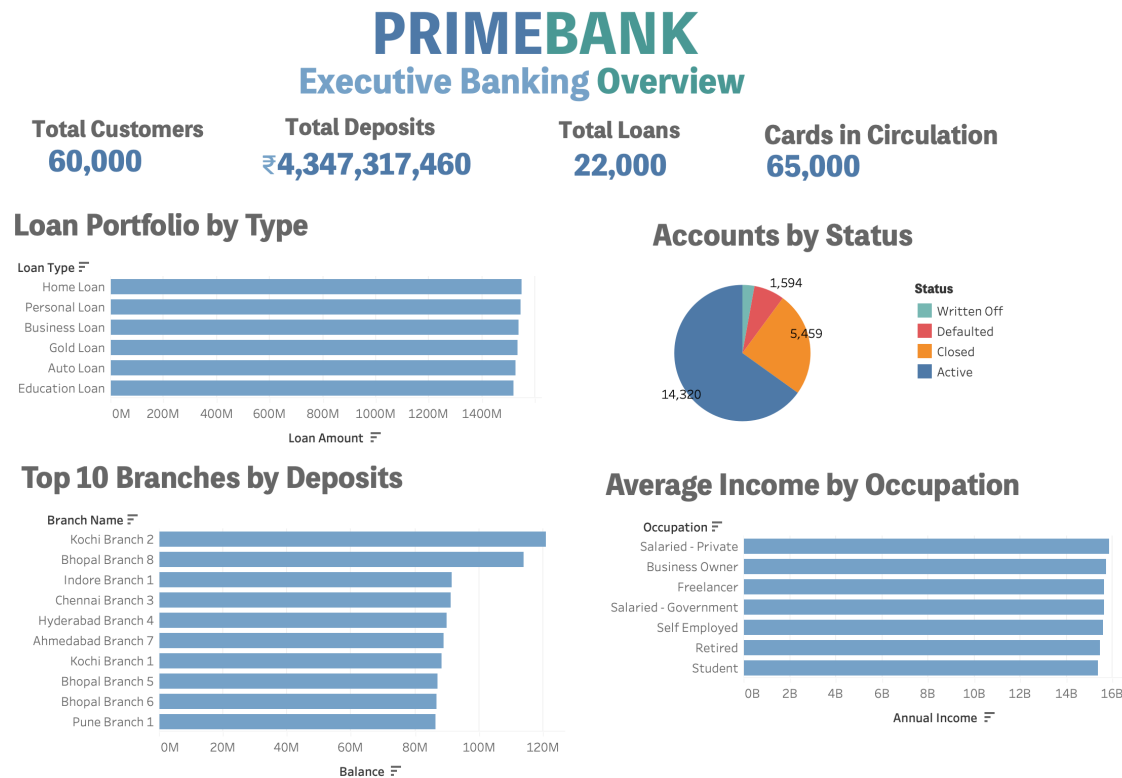


Figure — PRIMEBANK | Executive Banking Overview. Tableau Public dashboard view.

KEY TAKEAWAYS

- Establishes the bank-wide portfolio at a glance: customers, deposits, loans and cards.
- Compares loan mix, account status, leading branches and income by occupation.

PRIMEBANK | CUSTOMER 360

Customer base, demographics and credit profile.

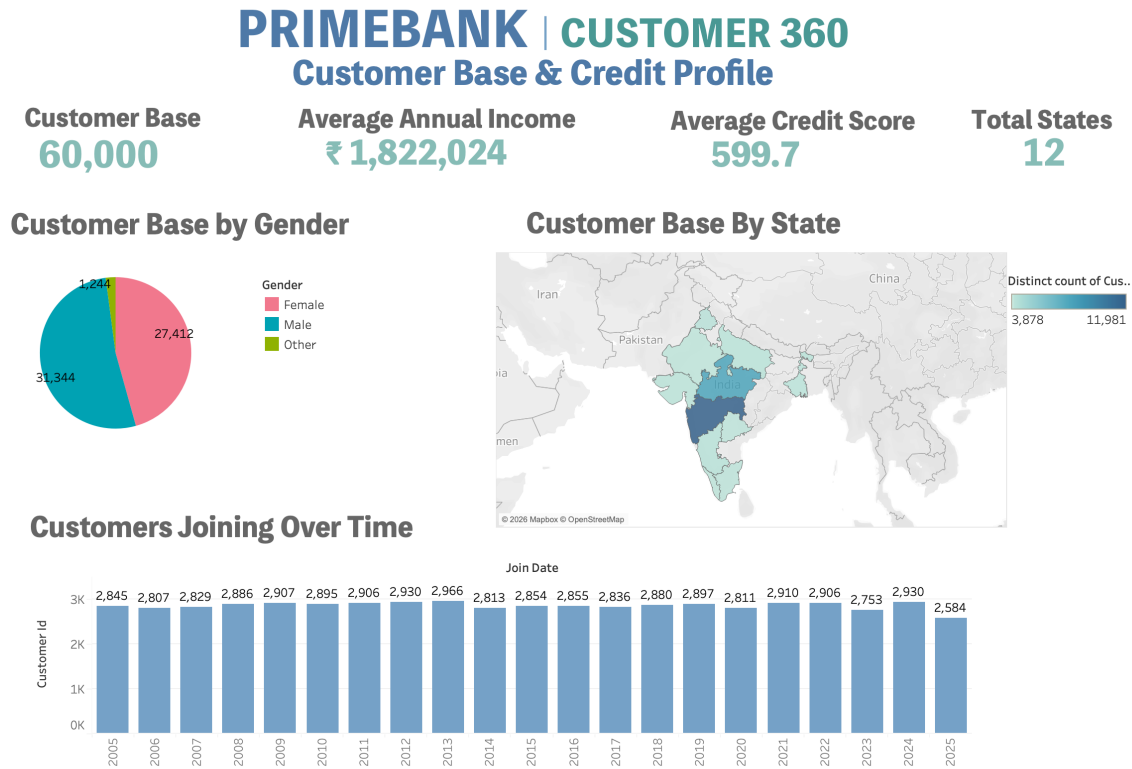


Figure — PRIMEBANK | CUSTOMER 360. Tableau Public dashboard view.

KEY TAKEAWAYS

- Summarizes customer scale, average income and average credit score.
- Adds gender/state distribution, acquisition trend and descriptive credit-profile views.

PRIMEBANK | Branch Performance

Branch network and operational performance.

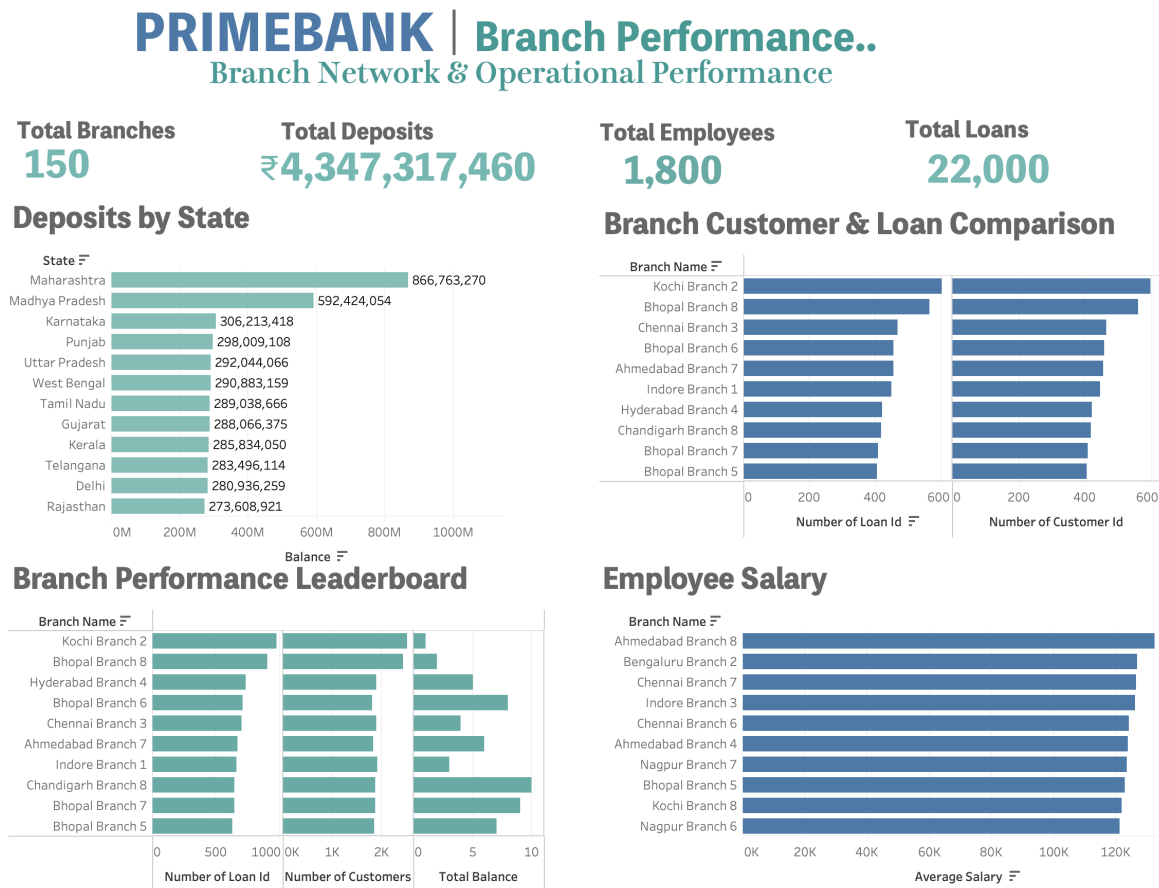


Figure — PRIMEBANK | Branch Performance. Tableau Public dashboard view.

KEY TAKEAWAYS

- Compares deposits by state and branch-level customer and loan activity.
- Includes a branch performance leaderboard and employee-salary view.

PRIMEBANK | MONEY MOVEMENT

Account health and transaction intelligence.

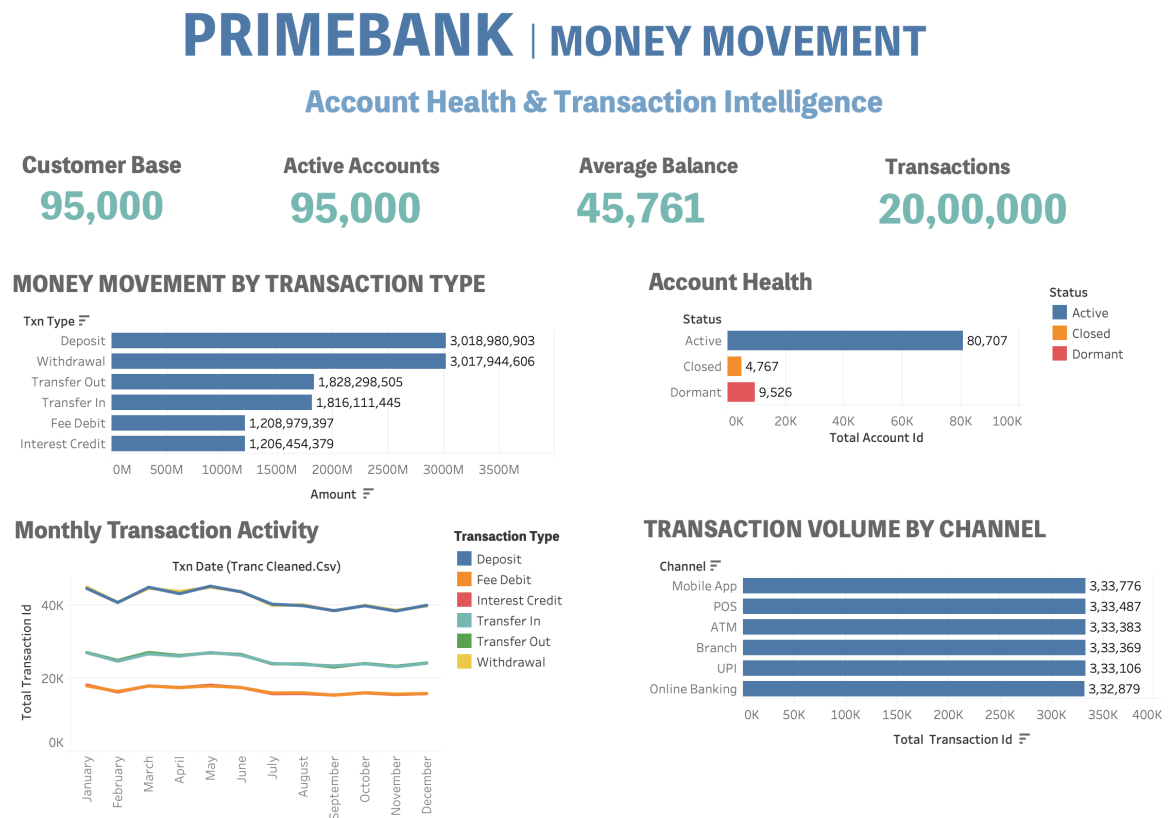


Figure — PRIMEBANK | MONEY MOVEMENT. Tableau Public dashboard view.

KEY TAKEAWAYS

- Shows transaction activity by type, channel and month alongside account health.
- Connects account-level records to broader money-movement patterns.

PRIMEBANK | LOAN COMMAND CENTER

Portfolio health, credit exposure and repayment performance.

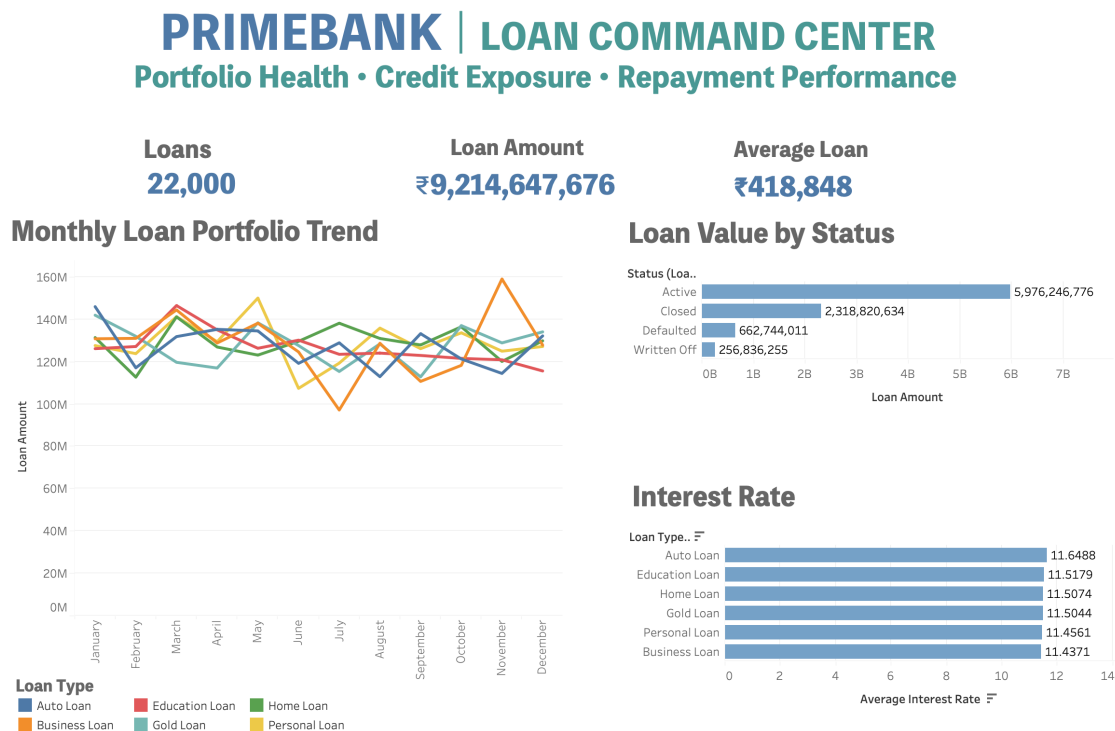


Figure — PRIMEBANK | LOAN COMMAND CENTER. Tableau Public dashboard view.

KEY TAKEAWAYS

- Tracks the 22,000-loan portfolio, portfolio amount and average loan size.
- Compares monthly movement, loan value by status and interest-rate patterns.

PRIMEBANK | FRAUD COMMAND CENTER

Card intelligence, merchant exposure and fraud monitoring.

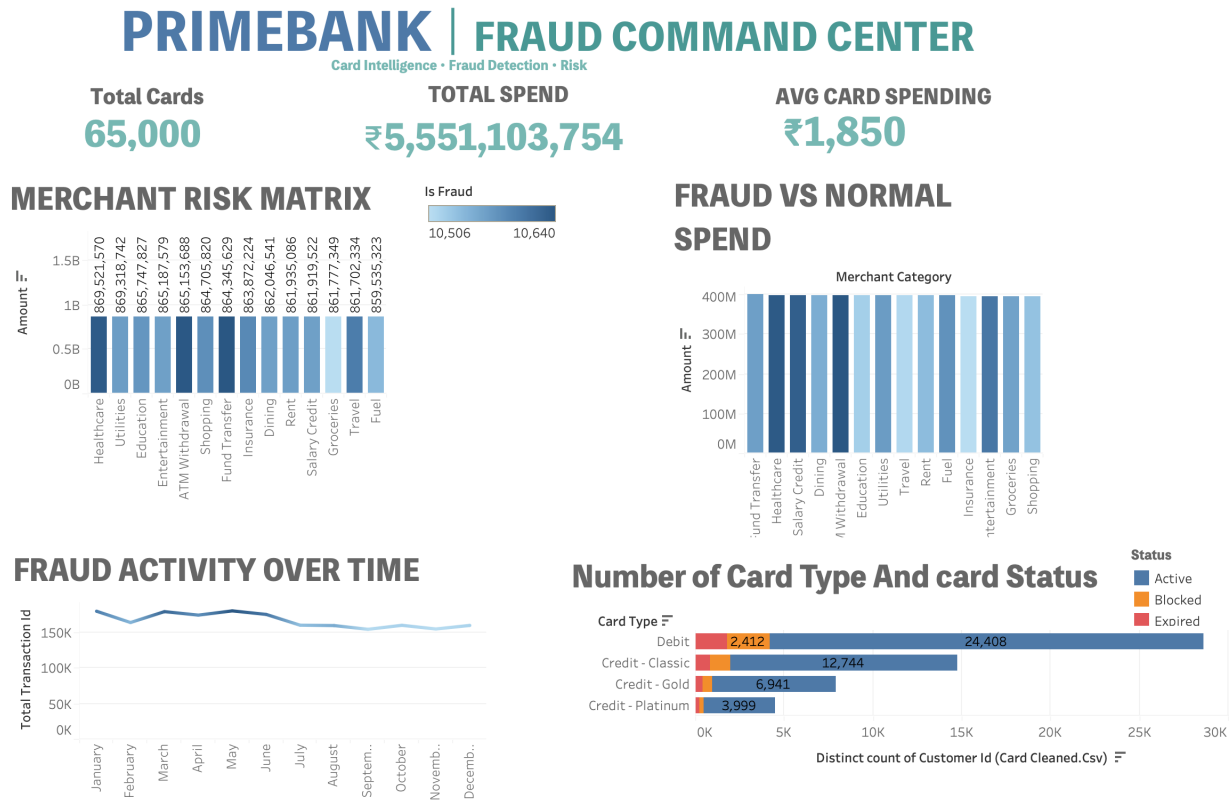


Figure — PRIMEBANK | FRAUD COMMAND CENTER. Tableau Public dashboard view.

KEY TAKEAWAYS

- Uses the 65,000-card portfolio as the card-domain anchor.
- Highlights merchant-category spending, fraud activity and card-status exposure.

PRIMEBANK | RISK COMMAND CENTER

Credit risk, default concentration and customer-support signals.

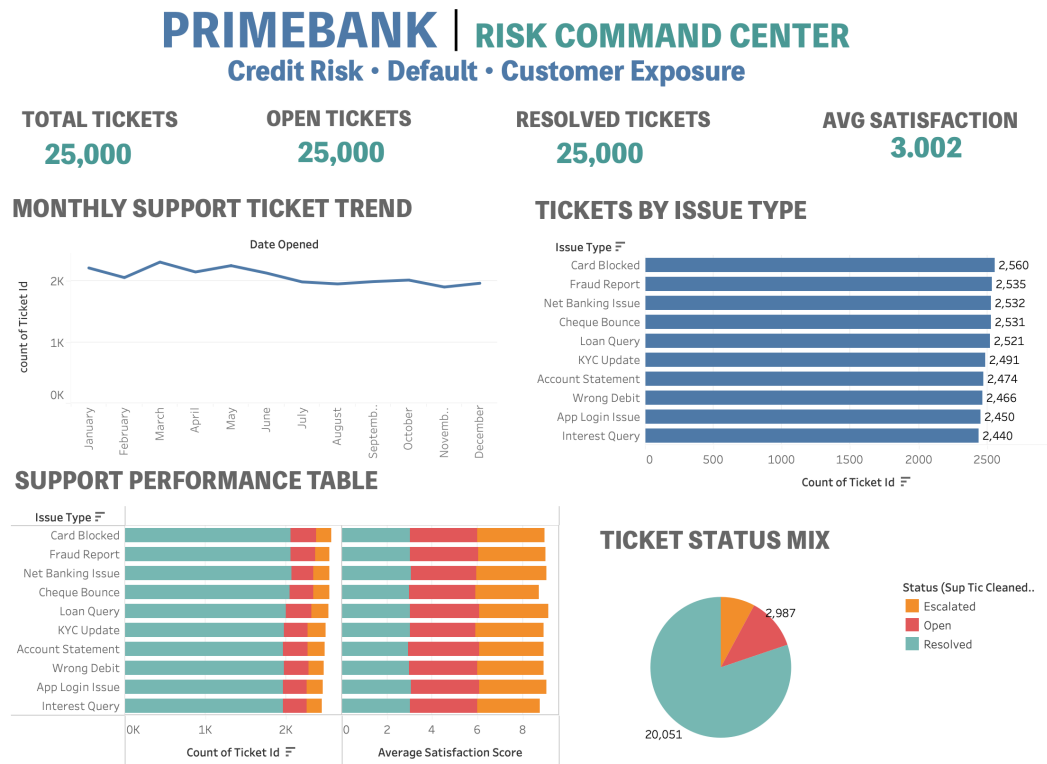


Figure — PRIMEBANK | RISK COMMAND CENTER. Tableau Public dashboard view.

KEY TAKEAWAYS

- Combines support-ticket trend, issue volume, ticket status and satisfaction detail.
- Provides a risk-oriented operational view alongside the project's credit and default analysis.

Cross-domain business insights

The analytical story that emerges when the domains are read together.

CUSTOMER	The 60,000-customer base supports segmentation by occupation, state, income and credit profile. Combining income and credit score gives a more useful descriptive view than demographics alone.
BRANCH	Deposit concentration varies by branch. The analysis also considers customer count, loan issuance, transaction volume and staffing, making branch performance broader than a single deposit ranking.
LOAN	The 22,000-loan portfolio is evaluated by type, status, default rate, branch, late-payment behavior and repayment economics. Repeated late payments are treated as an early-warning signal.
FRAUD	Card fraud is measured across 3,000,000 card transactions, with fraud and spending concentrated in specific merchant categories. The Fraud Command Center surfaces these patterns for monitoring.
SUPPORT	Customer support analysis covers 25,000 tickets, issue types, status and satisfaction. Average satisfaction is approximately 3.0/5, indicating a clear service-improvement opportunity.
RISK	Risk is concentrated rather than evenly distributed. Ahmedabad Branch 7 is highlighted as a high-default branch, while low credit score, high exposure and late-payment behavior form a useful descriptive risk screen.

Recommendations & action plan

Turning analytics into decisions.

AREA	RECOMMENDATION	EXPECTED VALUE
Credit Risk	Strengthen assessment for low-credit-score customers before new lending decisions.	Lower avoidable credit exposure
Loan Monitoring	Increase follow-up for customers with repeated late payments.	Earlier intervention
Branch Audit	Review high-default branches, especially Ahmedabad Branch 7, for lending and operational drivers.	Targeted risk reduction
Early Warning	Track repayment behavior trends as an early-warning layer.	Faster risk response
Fraud	Apply merchant-category-level fraud monitoring where exposure is concentrated.	Reduce fraud losses
Branch Strategy	Study deposit-leading branches and replicate effective practices.	Deposit growth
Support	Prioritize high-volume / low-satisfaction issue types for service improvement.	Higher customer satisfaction
Cross-sell	Prioritize relationship offers for customers already holding multiple accounts.	Higher product depth

Management principle

Do not spread attention evenly. Use the dashboards to identify concentrated exposure, then route the detailed customer, branch, loan, fraud or support records to the appropriate operational owner.

Technical challenges & what was learned

The strongest parts of the project are often the problems it had to solve.

JOIN FAN-OUT

Joining accounts, loans and transactions before aggregation can multiply rows and inflate sums/counts. Aggregate each domain independently, then join the compact results.

LARGE TABLES

Million-row transaction, card-transaction and loan-payment tables require efficient loading, indexing and careful query design.

METRIC SEMANTICS

A correct number can still be the wrong KPI if the field is mapped to the wrong business label. The Money Movement screenshot is a clear example.

DASHBOARD QA

Screenshots must be validated against source-table grain and denominator logic, especially for ratios such as fraud rate and default rate.

RISK INTERPRETATION

Low credit score alone is not treated as “risk”. The project combines multiple signals such as exposure and late-payment behavior for a more defensible descriptive screen.

A particularly important SQL pattern

```
WITH branch_deposits AS (...),
     branch_loans AS (...),
     branch_transactions AS (...)
SELECT ...
FROM branch_deposits
LEFT JOIN branch_loans ...
LEFT JOIN branch_transactions ...
ORDER BY ...;
```

Skills demonstrated

A recruiter-friendly translation of the technical work.

CATEGORY	EVIDENCE IN PROJECT
Python / Pandas	CSV ingestion, cleaning, type conversion, groupby/agg, merge, pd.cut, quantile-based segmentation
SQL / PostgreSQL	Joins, aggregation, CASE, FILTER, CTEs, subqueries, window functions, ranking
Data Modeling	Relational entities, primary/foreign-key thinking, one-to-many relationships
Analytics	Customer segmentation, branch performance, loan risk, fraud, transactions, support quality
Visualization	Seven Tableau Public dashboards, KPI design, trends, comparisons, operational scorecards
Data QA	Grain checks, KPI-label validation, denominator checks, join fan-out awareness
Business Communication	Insights translated into targeted recommendations and action areas

Portfolio positioning

This is not just a dashboard project. It demonstrates an end-to-end analyst workflow: understand the business problem → work with relational data → analyze with code → validate metrics → build decision-oriented dashboards → communicate actions.

Conclusion & project handoff

A complete analytical narrative, ready for portfolio use.

What the project achieves

- A 10-table banking data model spanning customers, accounts, branches, loans, cards, transactions and support.
- Python / Pandas analysis that prototypes and validates business questions.
- PostgreSQL / SQL analysis across customer, branch, account, loan, fraud and support domains.
- Seven current Tableau Public dashboard views that turn the analysis into a navigable management story.
- Risk-focused recommendations that connect findings to operational action.

Final publication checklist

- Reconcile the Executive deposit KPI against the final Tableau datasource.
- Correct the Money Movement KPI mappings (accounts vs customers; 2M vs 20M transactions).
- Ensure the Branch Performance KPI label matches the displayed employee metric.
- Add / validate Fraud Rate only with a full-transaction denominator.
- Keep the seventh page title aligned with the actual Risk Command Center screenshot.
- Use Vaibhav Singh as the report author everywhere.

Closing thought

The value of the project is not the volume of charts. It is the chain of reasoning from relational banking data to a small set of business decisions that a manager, branch lead, risk officer or support lead can act on.

Appendix — analysis coverage

Business questions represented in the Python and SQL work.

Customer Analysis

Occupation-wise average annual income • occupation-wise average credit score • credit-score bands • high income + low credit-score segments • city/state concentration • income vs credit score • account ownership • multiple accounts and balances • customers with loans + cards + multiple accounts

Branch Performance

Highest total deposits • highest number of customers • most loans issued • highest transaction volume • highest employee count • high deposits with relatively low loan activity

Account & Deposit Analysis

Most common account types • account types with highest total balance • highest average balance • branches with highest deposits • high customer count but relatively low deposits • multiple-account ownership • unusually high balances

Loan & Credit Risk

Loan portfolio by type • highest average loan amount • loan status distribution • default rate by loan type • highest branch default rate • repeated late payments • late-payment rate by loan type • principal / interest paid and outstanding • low credit + high exposure + late-payment customers

Card & Fraud

Card distribution by type • credit limits • merchant categories with highest spending • fraudulent vs legitimate transaction amount • unusual transaction activity / fraud exposure • high-value and high-volume accounts

Customer Support

Most frequent support issues • lowest satisfaction by issue • longest resolution time by issue • resolution time vs satisfaction • customer segments with most tickets / lowest satisfaction

Transaction Analysis

Highest transaction value by transaction type • most-used channels • monthly transaction volume and value trends